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Requirements Analysis Specification Document

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Students&Companies - Requirements Analysis Specification Document
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Contents

1 Introduction

The Requirement Analysis and Specification Document (RASD) aims to focus on the tasks needed to develop and implement an application, taking account of the requirements of the involved stakeholders and, analyzing and documenting also the application requirements. Then, in the second part of the document, there is a more formal definition of the requirements with the use of the Alloy language. In general this document is meant for developers tasked with the implementation of the system and also for all the other entities involved in validation and managing of the project.

1.1 Purpose

Recently job market becoming more and more competitive. Accordingly students face significant challenges in finding suitable job opportunities and internships with respect to their experiences, skills and most importantly their interest and their career goals. Simultaneously, companies find it hard who with fit their need and requirements as well as their work culture. The Students&Companies (S&C) platform addresses this need by streamlining and enhancing the process of matching students with relevant internship opportunities. By analyzing the experiences, skills and attitudes of students listed in their CVs and comparing them with the specific requirements and perks of available roles provided by the companies, S&C connects students with positions that suit them well.

The platform helps students easily find internship opportunities by collecting detailed information about available internships, student resumes, skills, and preferences. This allows students to search for companies and roles that match their interests through S&C's search system, giving them more control over finding an internship. Additionally, S&C assists companies by managing interview schedules and assessments for students who apply for specific roles, making the hiring process smoother and helping both sides make better decisions. Finally, S&C offers tools to track internships by gathering student feedback and complaints, providing insights to improve future programs. Overall, S&C aims to connect students and companies, making it easier and more transparent for students to start their careers.

This document focuses on the Requirements Analysis and Specification Document (RASD) of the system and describes the main goals, the domain assumptions, the scenarios which may happen, the uses cases, the list of functional and non-functional requirements which system should fulfill, and finally the diagrams to visualize the interactions between components and performance of the system.

1.1.1 Goals

Table 1: Goals Definition

Goals	Description
G1	S&C allows students to manage (create, upload, edit) their resume
G2	S&C allows students to see available internships (e.g., by location, industry)
G3	S&C allows students to initiate the search process and submit applications directly
G4	S&C allows students to track the status (reviewed, shortlisted, selected) of their internship applications
G5	S&C allows students to raise complaints or concerns regarding an internship experience
G6	S&C allows companies to manage (create, edit, publish, delete) internship listings
G7	S&C allows companies to browse and search through student profiles (CVs) applied for the internship (e.g., by location, industry)
G8	S&C allows companies to review incoming applications
G9	S&C allows companies the ability to shortlist candidates
G10	S&C allows companies communicate with applicants
G11	S&C allows companies to provide updates on their application status of incoming applications
G12	S&C allows companies to manage (create, edit, send) questionnaire to applicants
G13	S&C allows companies to manage (scheduling and conducting) interviews with applicants
G14	S&C allows universities to track application processes
G15	S&C allows universities to monitor internship execution
G16	S&C allows universities to collect complaints

1.2 Scope

The Students & Companies (S&C) platform connects and facilitate matching university students want to find internships with companies providing those opportunities. The platform will provide abilities for students to manage their profiles, apply for internships, and track their application status. At the same time, it will allow companies to post internships offerings, review applications, and communicate with potential candidates. The platform also includes an access for universities to monitor internship activities and collect compliance.

To initiate application process, an application has been defined through which the user can do the following:

Student Features

- Students must register to system with their credentials.

- Student must log in to his/her account. Then, students can create, upload, and edit their resumes.
- Then, students can view available internships and filter them based on various criteria.
- After selecting desired internships, students can initiate applications for internships and track the status of their applications (e.g., reviewed, shortlisted, selected).
- Furthermore, students can raise complaints or concerns regarding their internship experiences.

Company Features

- Companies must register to system with their credentials.
- Companies must log in to their account, and then, an create, edit, publish, and delete internship listings with all the necessary information (e.g. requirements, details of the job, ...)
- Then, for each internship posting, companies can browse and search through student profiles (CVs) that have applied for their internships.
- After receiving applications, companies can review incoming applications and shortlist candidates based on their qualifications.
- Then, companies can communicate with applicants and provide updates on application statuses.
- Companies can schedule and conduct interviews with shortlisted candidates and manage questionnaires for applicants.
- Finally, companies can decide which applicants are suitable for the positions based on interview and questionnaire.

Universities Features

- universities must register and login to their account.
- Then, universities can monitor the application processes of their students.
- Also, universities can oversee the execution of internships to ensure compliance and quality.

1.2.1 World Phenomena

Table 2: World Phenomena

World Phenomena	Description
WP1	Students actively look for internship opportunities based on their interest and skills and knowledge
WP2	Students prepare CVs
WP3	Companies offer various internships with specific requirements, expectations, and work cultures, which are of interest to potential student applicants.
WP4	Companies review students CVs
WP5	Companies decision on short listing candidates
WP6	Companies decision on acceptance/rejection of candidates
WP7	Students accept/reject the offer of internship
WP8	Students start the internship

1.2.2 Shared Phenomena

Table 3: Shared Phenomena

Shared phenomena	Description	Control
SP1	Students create, upload, and edit their resumes within the system to apply for internships	World
SP2	Companies create, publish, edit, and delete internship listings through the platform	World
SP3	System shows the available internships	System
SP4	Students use the system's search and filtering functions to explore available internship opportunities	World
SP5	Students apply to the internship through the system	World
SP6	Companies search and filter applicants through the system	World
SP7	Companies manage (create, edit, publish, delete) questionnaire through the system	World
SP8	Companies schedule a meeting through the system	World
SP9	Students are notified about a change in applications	System
SP10	System shows all the students applied for an internship	System
SP11	System notifies the universities about the status and process of internship	System
SP12	System notifies the companies about whether the internship accepted/rejected	System

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

1.3.2 Acronyms

1.3.3 Abbreviations

Table 4: Abbreviations

Abbreviations	Description
G	Goal
R	Requirement
C	Component
WP	World Phenomena
SP	Shared Phenomena

1.4 Revision history

1.5 Reference Documents

- Specification Document: "Assignment RDD AY 2024-2025.pdf"
- Course Slides

1.6 Document Structure

Section 1 Introduces the problem and the purpose of the project and defining the scope of the Students&Companies. Describe the specifications such as the definitions, acronyms, abbreviations, revision history, and references. As well as introducing the goals, world and shared phenomena of the software.
Section 2 Identifying key scenarios and detailing the software's primary features through class diagrams and state charts. The user characteristics section describes the types of actors interacting with the application, while the product function subsection outlines its functionalities. Finally, domain assumptions are established.
Section 3 The main part of the project which introduces interface requirements such as user interface, hardware interface, software interface, and communication interfaces. Presenting the functional requirements that are shown by use case diagrams and sequence diagrams. Then requirements are mapped to use cases.
Section 4 Using Alloy language for analyzing the system and brief comments for clarifying the Alloy codes.
Section 5 Shows how much time is spent on this project.
Section 6 Contains the references

2 Overall Description

2.1 Product perspective

2.1.1 Scenarios

Student Uploads CV Mia is sitting at her desk at the university library, she is excited to showcase her skills, capabilities, and experience on the S&C platform. She navigates to her profile page, where she finds an option to upload her CV. Mia selects "Add CV", which opens a file explorer window. With precision, she locates her CV file within the organized "Career" folder on her desktop. She clicks to upload the document. The platform start the upload process, automatically checking the document format to ensure compatibility. A progress indicator shows the confirmation of the upload, and once completed, Mia can review her uploaded CV to confirm accuracy and ensure it aligns with her career goals and information.

Company Posts Internship Alex is a representative of company called BigTech and working in hiring manage department. She has been notified by her boss that a new intern is required for R&D department. Accordingly, Alex is preparing to post a new internship. He logs into the platform, selects "Post Internship," and begins to fill out the details. He describes the responsibilities, the technical skills required, and any perks like mentorship or training programs. After adding all the information, Alex hits "Submit." Within moments, Mia and other students can now view the new opportunity and consider applying.

Viewing internship opportunities On the S&C platform, Emma uses the search feature to find internships by entering keywords like "machine learning," "AI," and "data analysis." She filters results by location, seeking options close to her university campus, and sorts the list by the latest postings. After browsing several listings, Emma finds the TechVision Corp internship. she clicks in to learn more, reading through the description to see if it aligns with her career aspirations. She reviews the details of the position, compares it with other internships, and decides it's a good match.

Student Initiates Application After reading through the details, Mia feels Alex's internship is a perfect match for her goals. She clicks "Apply," and her CV is immediately sent to Alex's company profile. The platform confirms her application, and Mia's dashboard updates to show the internship as "Applied." She's thrilled to take her first step towards the role.

Company views all the applicants for internship posting Sarah, an HR manager at TechVision Corp, has posted an internship for a machine learning role on the S&C platform. After several days, she has received numerous applications. Sarah logs into the S&C platform and navigates to the "Applications" dashboard for her internship posting. The platform provides a list of applicants, along with their CVs. Sarah uses the platform's filtering options to shortlist applicants based on criteria such as relevant skills and project experience, enabling her to focus only on the most qualified candidates.

Scheduling Interviews by company's hiring manager After reviewing the applications, Sarah decides that Emma and a few other candidates are well-suited for the role. She wants to set up interviews to get to know them better. Sarah selects Emma's profile and clicks "Schedule Interview" on the S&C platform. A calendar view appears, showing Emma's availability based on her provided schedule, along with suggested time slots to minimize conflicts. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting.

Creating a Questionnaire for Candidate Assessment To streamline the interview process, TechVision Corp uses a structured questionnaire to gauge candidates' technical knowledge and problem-solving skills. Sarah logs into the S&C platform and navigates to the "Assessment Tools" section. Here, she finds options to create a new questionnaire or select pre-made templates relevant to her internship role. Since she wants to evaluate specific skills related to TechVision's project requirements, Sarah opts to create a new questionnaire from scratch. She decides that the questionnaire should cover three main areas: technical knowledge (multiple-choice), coding skills (coding challenges), and scenario-based problem-solving (short-answer questions). After building the questionnaire, Sarah reviews each question for clarity, relevance, and alignment with the internship's objectives. She configures the scoring criteria for each section. After the questionnaire has been completed, she send and publish this questionnaire to all the applicants applied for the internship position.

Answering to questionnaire Emma logs into her S&C account and sees a notification on her dashboard about a new task. She clicks on the alert, which opens a detailed view of her application progress. A message from TechVision Corp indicates that she needs to complete a pre-interview questionnaire within three days. Emma also receives an email reminder with a direct link to the questionnaire. Emma clicks on the link in her dashboard, which takes her to the "Assessment" section of her application profile. Here, she sees an overview of the questionnaire, including the expected completion time. After completing all sections, the platform allows Emma to review her responses. She takes a final look to confirm that she hasn't missed any questions and feels satisfied with her answers. Emma clicks "Submit," and the S&C platform displays a confirmation message, assuring her that her responses have been successfully sent to TechVision Corp's HR team.

Finalizing the Evaluation and Making a Decision After interviewing multiple candidates, Sarah accesses the S&C platform to review her evaluation notes, assessment scores, and feedback from each candidate's questionnaire. After finalizing her decision, Sarah updates Emma's application status to "Accepted" on the S&C platform. This change automatically notifies Emma, informing her that she has been selected for the internship and can review her offer details.

University Internship Monitoring The university representative, Professor Green, logs into the S&C platform. Upon accessing the university dashboard, she sees an overview of all ongoing internships involving students from her university. Professor Green selects the "Internship Monitoring" tab, where she views a list of students currently interning, along with detailed internship information such as company names, role titles, start and end dates, and supervisors. Each entry shows the student's progress and any recorded evaluations or feedback from the company.

Student Complaint Submission Mia, a student currently in an internship, encounters issues regarding her working hours. She logs into her S&C profile, navigates to "Internship Support," and selects "Submit Complaint.". Mia fills in the complaint form, specifying the issue and selecting "Excessive Work Hours" from a list of predefined categories. She adds further comments to explain her situation and attaches relevant evidence, such as email correspondence with her university supervisor.

Receiving Complaint By University Professor Green, who is responsible for overseeing student internships at the university, wants to check if any students have submitted complaints regarding their internships. Using the S&C platform, she can efficiently access and manage all student complaints in one place. Professor Green logs into the S&C platform and navigates to the "Complaints" section in her dashboard. Here, she has a dedicated view displaying all submitted complaints from students at her university. On this page, she sees a list of complaints categorized by student names, company names, submission dates, and complaint categories (e.g., "Work Hours," "Workplace Environment,"). Each entry also has a brief description of the issue for quick reference. To prioritize her review, Professor Green uses filtering options to sort complaints by severity, date submitted, or status (e.g., "New," "In Progress," "Resolved"). This helps her quickly identify urgent issues.

2.1.2 Class Diagram

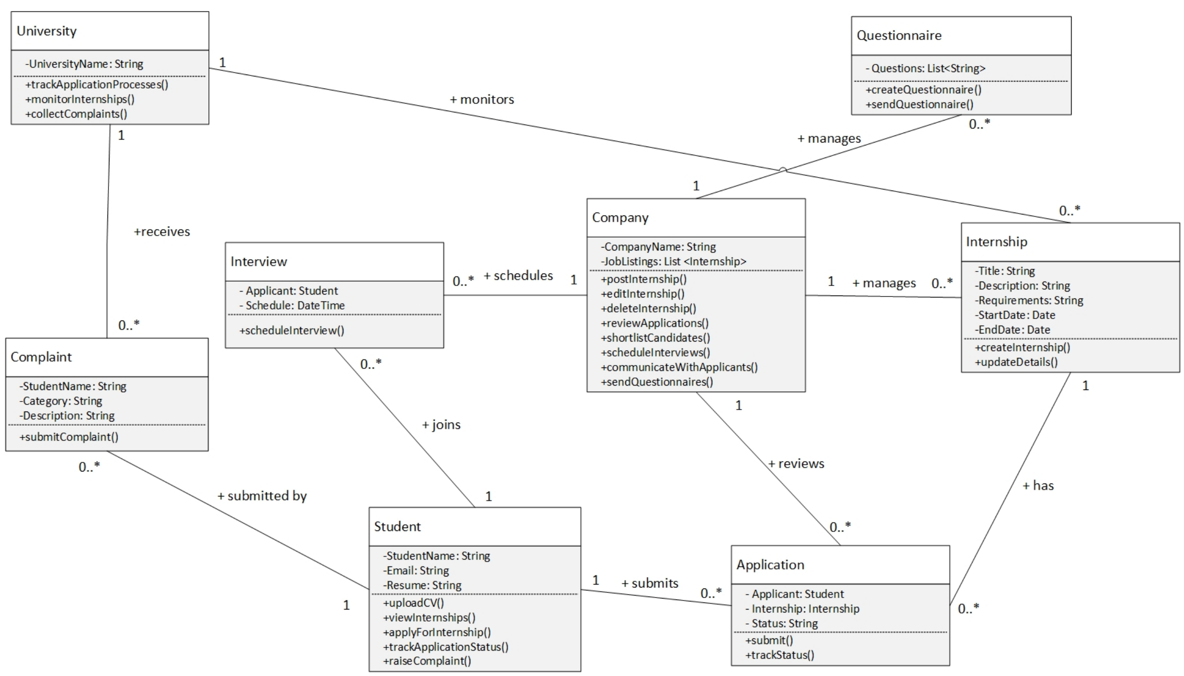

Figure 1: Class Diagram

2.1.3 State Diagram

2.2 Product Functions

2.3 User Characteristics

2.3.1 Actors

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

2.4.2 Dependencies

2.4.3 Constraints

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

3.1.3 Software Interfaces

3.1.4 Communication Interfaces

3.2 Functional Requirements

3.2.1 Use Case Diagram

3.2.2 Use Cases

3.2.3 Sequence Diagram

3.2.4 Goal Mapping on Requirements

3.3 Performance Requirements

3.4 Design Constraints

3.4.1 Standard Compliance

3.4.2 Hardware Limitations

3.4.3 Any Other Constraint

3.5 Software System Attributes

3.5.1 Reliability

3.5.2 Availability

3.5.3 Security

3.5.4 Maintainability

3.5.5 Portability

4 Formal Analysis using Alloy

4.1 What we want to model?

4.2 Alloy Model

4.3 Alloy Generated Worlds

5 Effort Spent

References